Project Guidelines

Due Monday by 11:59pm

Points 100

Submitting a file upload

Available Jan 28 at 12am - Mar 17 at 11:59pm

Minimum 10 pages. Max 12 pages. The project proposal report should be written using latex or overleaf. The proposal should contain **title, abstract, introduction, problem statement**, related work, **proposed solution, proposed list of experiments, experiment set-up and data set details, proposed eval metrics**, detailed project execution timeline, references. The data sets used must be open source.

You are welcomed to use open source code and modify it to meet your project needs. The source code repository must be cited in the project proposal.

You are welcome to use Gen-AI to help you understand open source code/ repositories but using gen-AI solely to generate your project proposal or generating complete code for your project will result in you failing the course.

Accelerating Inverse Lithography Technology (ILT)

Accelerating Inverse Lithography Technology Using Depthwise Separable CNNs

Evaluation on the LithoBench MetalSet Dataset

Abstract

As Moore's Law nears 3nm nodes, optical lithography struggles with diffraction [3]. Traditional correction methods (OPC, iterative ILT) are computationally intensive for complex layouts [4][5]. Machine learning approaches, using neural networks and frameworks like LithoBench, accelerate mask optimization by learning the layout-to-mask mapping and reducing physics simulations. However, the standard U-Net architectures used with LithoBench have high computational overhead.

This project proposes a more efficient NeuralILT approach using depthwise separable convolutions (like in MobileNet) to reduce parameters and FLOPs. Using the LithoBench MetalSet, the model aims to lower computational cost while maintaining accuracy for advanced nodes. Performance will be compared against the standard U-Net baseline using MSE, SSIM, and EPE. The expected outcome is a lightweight NeuralILT model that improves efficiency without sacrificing mask fidelity.

Introduction

As semiconductor technology scales, lithographic complexity increases, necessitating mask correction to address patterning distortions. While early methods like Optical Proximity Correction (OPC) are insufficient for modern designs [4], Inverse Lithography Technology (ILT) significantly improves fidelity [5] with pixel-based optimization. However, ILT is computationally expensive due to repeated physics simulations. NeuralILT uses deep, resource-intensive CNNs like U-Net to accelerate mask synthesis by learning target-to-mask mappings, reducing the need for extensive simulations.

This project proposes a more efficient NeuralILT approach using depthwise separable convolutions to reduce parameters and floating-point operations while maintaining mask prediction accuracy. Using the LithoBench MetalSet dataset, the project will evaluate if this efficient CNN architecture can accelerate NeuralILT while preserving the fidelity required for advanced semiconductor manufacturing.

Problem Statement

Although NeuralILT has significantly accelerated the traditional inverse lithography optimization process, existing implementations remain computationally expensive when applied to large-scale semiconductor layouts [1]. The standard U-Net architecture commonly used with the LithoBench MetalSet dataset can efficiently process individual layout tiles, but scaling these models to full-chip optimization requires substantial computational resources and memory [1]. This computational burden arises from the large number of convolutional kernels and parameters required to capture the complex curvilinear mask patterns present in advanced lithography datasets.

As semiconductor technology shrinks, the high computational cost and parameter count of inverse lithography (NeuralILT) models pose deployment challenges. This project will investigate using depthwise separable convolutions to redesign the NeuralILT architecture. The goal is to reduce computational cost and complexity while maintaining mask prediction accuracy on the LithoBench MetalSet dataset.

Related Work

Computational lithography research has increasingly focused on balancing modeling accuracy with computational efficiency as semiconductor technology scales to smaller nodes [7]. Traditional lithography modeling relies on detailed physical simulations of optical imaging, resist exposure, and etching processes, which can achieve high accuracy but are computationally expensive for full-chip optimization. Early machine learning methods such as Support Vector Machines and decision trees were introduced to improve lithography simulation and defect detection tasks, demonstrating that data-driven models could approximate complex lithographic behavior without solving the underlying physical equations directly [8,9].

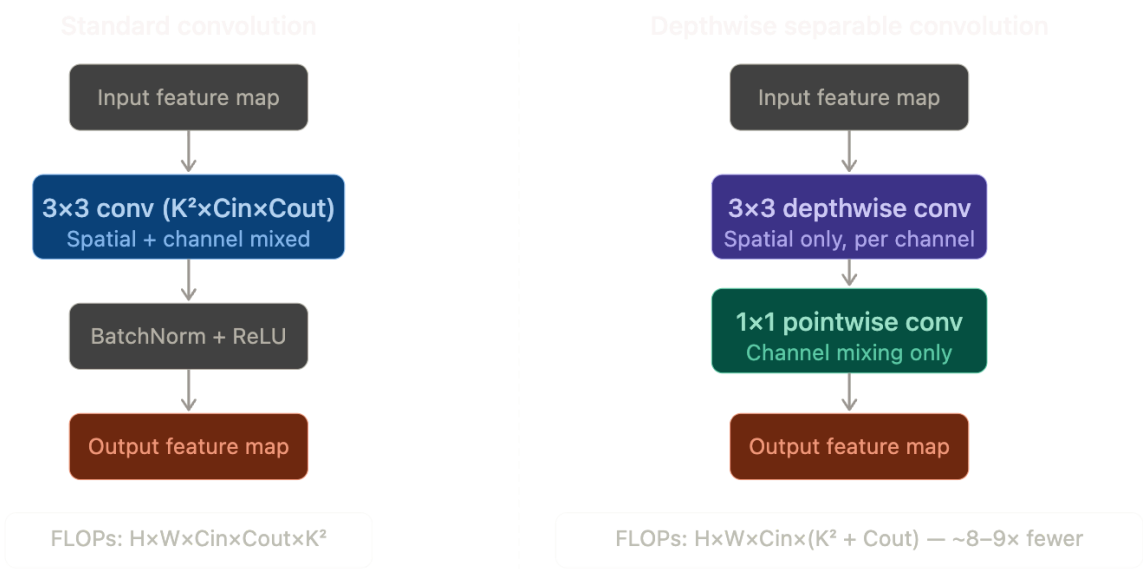
Deep learning, including CNNs, fully convolutional networks, and generative models like LithoGAN and TEMPO, has recently accelerated lithography modeling, simulation, and mask optimization, achieving accuracy comparable to rigorous simulations at a lower computational cost [10–16]. These techniques have also been applied to Inverse Lithography Technology by learning target-to-mask mappings [17, 18]. Model-driven deep learning further improves reliability by incorporating physical imaging models [19–21]. However, the reliance on computationally expensive CNN architectures motivates exploring more efficient designs, such as depthwise separable convolution, for NeuralILT models.

Proposed Solution

This project proposes a lightweight NeuralILT architecture using depthwise separable convolutions within a CNN, trained on the LithoBench MetalSet dataset. The goal is to significantly reduce computational cost and model complexity while maintaining mask prediction accuracy for advanced semiconductor manufacturing.

Current NeuralILT models, often U-Net-like encoder–decoders, are computationally expensive for large datasets and full-chip optimization. In this project, the baseline model is a standard U-Net encoder–decoder architecture using 3×3 convolution layers and skip connections, similar to those used in LithoBench NeuralILT experiments [1]. The proposed solution replaces standard convolutions with depthwise separable convolutions [6], which split the operation into independent depthwise (spatial filtering) and pointwise (channel mixing) stages. This separation drastically reduces parameters and FLOPS, as demonstrated by architectures like MobileNet.

The architecture is expected to significantly reduce FLOPs due to replacing standard convolutions with depthwise separable convolutions, whose computational complexity is shown in Computational Cost of Convolution Operations.



Computational Cost of Convolution Operations

We compare standard and depthwise separable convolution computational complexity, measured in FLOPs, to motivate the proposed architecture change.Standard Convolution FLOPs
For a standard convolution layer, the number of floating-point operations is:

$$FLOPs_{std} = H \times W \times C_{in} \times C_{out} \times K^2$$

Where:

- (H, W) = height and width of the feature map
- (C_{in}) = number of input channels
- (C_{out}) = number of output channels
- (K) = kernel size (e.g., $(K = 3)$ for a (3×3) convolution)

In this operation, spatial filtering and channel mixing occur simultaneously, which leads to a large number of parameters and high computational cost.

Depthwise Convolution FLOPs

Depthwise convolution applies a spatial filter independently to each input channel:

$$FLOPs_{dw} = H \times W \times C_{in} \times K^2$$

This operation captures local spatial patterns in each feature map while avoiding channel mixing.

Pointwise Convolution FLOPs

Pointwise convolution uses a (1×1) kernel to combine information across channels:

$$FLOPs_{pw} = H \times W \times C_{in} \times C_{out}$$

This stage performs channel mixing after the spatial filtering step.

Total FLOPs for Depthwise Separable Convolution

Combining both operations results in the total computational cost:

$$FLOPs_{sep} = H \times W \times (C_{in} \times K^2 + C_{in} \times C_{out})$$

Comparing the two formulations shows that depthwise separable convolution removes the $C_{in} \times C_{out} \times K^2$ term present in standard convolution. Instead, the computation is split into spatial filtering ($CinK^2$) and channel mixing ($CinCout$), which significantly reduces the total number of floating-point operations, particularly when $Cout$ and K are large.

Depthwise separable convolution reduces computational cost significantly compared to standard convolution while maintaining feature extraction. This makes it suitable for improving NeuralILT efficiency on the LithoBench dataset. A 3x3 kernel with Cout = 64 can reduce FLOPs by approximately an order of magnitude depending on channel dimensions.

Expected Outcomes and Hypothesis

The NeuralILT architecture, leveraging the properties of depthwise separable convolution (DS-Conv) and lightweight CNNs like MobileNet [6], aims to boost computational efficiency while preserving mask prediction accuracy.

Key hypotheses include:

- **Computational and Inference Efficiency:** The DS-Conv will significantly reduce FLOPs and parameter count compared to the standard U-Net baseline, making it better for large-scale synthesis. This reduced complexity is expected to yield 2–4× faster GPU inference time, even considering memory bandwidth limits.
- **Mask Prediction Accuracy:** DS-Conv should retain high representational capability, with SSIM scores potentially within 2–5% of the baseline, based on the representational capability of depthwise separable CNNs.
- **Generalization:** With fewer parameters, the model may reduce overfitting on MetalSet, generalizing similarly or better than the baseline on the out-of-distribution StdMetal dataset.

These hypotheses will be validated by the proposed experiments and metrics.

Training Strategy

We will train a model using supervised learning with the LithoBench dataset's ground truth mask patterns. The neural network will learn to map the target layout to the optimized mask pattern from inverse lithography technology (ILT) [1]. We will use the MetalSet subset, taking target layout tiles as input and ground truth mask images as labels. The goal is accurate prediction of optimized mask patterns, addressing the complex metal routing and strong optical proximity effects. During training, we will use Mean Squared Error (MSE) to measure the pixel-level difference between the predicted mask output and the ground truth mask intensities.

$$L_{MSE} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2$$

Where:

- (y_i) represents the ground truth mask value
- ($\hat{y}_i$) represents the predicted mask value
- (N) represents the total number of pixels in the image

We will use Structural Similarity Index (SSIM) alongside pixel-wise loss to ensure structural fidelity between predicted and ground truth lithography masks.

Training will employ PyTorch, a GPU, and the Adam optimizer with mini-batch training. To ensure a fair comparison, the depthwise separable architecture will use the same dataset and training setup as the baseline NeuralILT-style model.

Evaluation will use the MetalSet testing subset and the StdMetal dataset to assess generalization.

Proposed List of Experiments

The experiments aim to evaluate whether replacing standard convolution layers with depthwise separable convolutions improves the computational efficiency of NeuralILT models while maintaining mask prediction accuracy.

Experiment 1: Baseline NeuralILT Model

A baseline model based on a standard U-Net encoder–decoder architecture with $3\#3$ convolution layers and skip connections will be trained using the LithoBench MetalSet dataset[1].

- Input: Target layout tiles
- Output: Optimized mask tiles
- Training: Supervised learning using ground truth masks

This experiment establishes baseline performance for both mask prediction accuracy and computational cost.

Experiment 2: Depthwise Separable Convolution Model

The second experiment replaces the standard convolution layers in the baseline architecture with depthwise separable convolution layers. The training setup, optimizer, dataset split, and hyperparameters will remain identical to Experiment 1 to ensure a fair comparison.

Experiment 3: Performance Comparison

The baseline and proposed models will be compared using both accuracy and computational efficiency metrics.

Mask Fidelity Metrics

- Edge Placement Error (EPE): distance between predicted mask edges and ground truth edges
- Structural Similarity Index (SSIM): structural similarity between predicted and ground truth masks

Computational Efficiency Metrics

- FLOPs (floating-point operations)
- Parameter count
- GPU inference runtime
- Memory usage

Experiment 4: Generalization Evaluation

Both models will be evaluated on the StdMetal dataset to measure generalization performance on layouts outside the MetalSet training distribution [1].

Experimental Objective

These experiments aim to determine whether the proposed depthwise separable convolution architecture can:

- Reduce computational cost and model complexity
- Maintain acceptable mask prediction accuracy
- Improve the scalability of NeuralILT models for large-scale lithography workflows

Experiment Setup and Dataset Details

This project will use the open-source LithoBench dataset [1][2] for experiments. LithoBench provides layout–mask pairs to train and evaluate neural networks for inverse lithography technology (ILT). The data is based on ICCAD-13 CAD contest benchmarks, which define design rules and simulate realistic optical proximity effects for semiconductor manufacturing.

The dataset contains over 120,000 layout tiles and is divided into four subsets:

Dataset	**Number of Tiles**	**Description**
MetalSet	16,472	Complex metal routing patterns with strong optical proximity effects
ViaSet	116,415	Simpler via patterns with relatively isolated structures
StdMetal	271	Standard metal layouts used for evaluating model generalization
StdContact	328	Standard contact patterns used for cross-domain testing

The NeuralILT models will primarily be trained on the MetalSet dataset due to its complex optical proximity effects. Model generalization will be evaluated using the StdMetal dataset for unseen layouts. StdContact may be used for additional cross-domain testing on structurally different contact layer patterns. The simpler ViaSet subset will help analyze model behavior under less complex optical conditions.

Data Preprocessing and Augmentation

To ensure data quality and a fair comparison between the baseline U-Net and the proposed Depthwise Separable Convolution (DS-Conv) models, all raw tiles from the LithoBench dataset undergo a standardized preprocessing pipeline.Preprocessing Steps

Pixel intensities for layout input and ground truth mask labels are normalized to $[0, 1]$, by dividing by 255. This normalization ensures stable gradient flow and consistent convergence. The 256-pixel LithoBench dataset tiles are loaded as single-channel grayscale images using PyTorch's Dataset/DataLoader, with shuffling enabled for training and disabled for validation/testing.

Dataset Split and Evaluation Sets

The 16,472 tiles of the MetalSet dataset are split with a fixed random seed to ensure reproducible results:

Set	Proportion	Number of Tiles	Purpose
Training	80%	13,178	Model training
Validation	10%	1,647	Hyperparameter tuning and model selection
Test	10%	1,647	Final performance evaluation

The StdMetal (271 tiles) and StdContact (328 tiles) sets are excluded and reserved solely for final evaluation. To boost model robustness during training, physically meaningful data augmentation (horizontal flips, vertical flips, 90-degree rotations) is stochastically applied, but no photometric transformations are used due to the physical significance of pixel values. This identical data processing pipeline is used for both the proposed DS-Conv model and the baseline U-Net to ensure a fair performance comparison.

Environment

Hardware and Software for Inverse Lithography Project

Experiments will use a GPU-enabled environment (Google Colab with NVIDIA T4 or G4) for accelerated training and inference of neural networks used in NeuralILT models.

The software stack includes Python 3, PyTorch for deep learning, NumPy, SciPy, and scikit-image (for metrics like SSIM). This environment enables efficient training of both the baseline and proposed depthwise separable convolution models, allowing accurate evaluation of lithography mask prediction.

Proposed Evaluation Metrics

Model performance will be compared to the baseline NeuralILT using accuracy and computational efficiency metrics.

Accuracy Metrics

Edge Placement Error (EPE)
EPE is a key metric in computational lithography, measuring the distance between predicted and ground truth mask edges. Lower EPE indicates a more accurate mask reproduction of the intended layout [1].

Structural Similarity Index (SSIM)
SSIM assesses the perceptual similarity between predicted and ground truth mask images, focusing on structural elements like edges and contrast. It is useful for evaluating mask fidelity. SSIM ranges from **1** (identical) to **0** (no structural similarity); higher values indicate better similarity.

Computational Efficiency Metrics

We will evaluate the model's computational efficiency and complexity using several metrics:

- Parameter Count: Measures total trainable parameters; depthwise separable convolution is expected to reduce this significantly.
- FLOPs: Measures floating-point operations to estimate computational cost; the architecture aims to reduce FLOPs via depthwise separable layers.
- Inference Runtime: Measures time for mask prediction on the GPU.
- Memory Usage: Measures GPU memory consumption during inference.

These metrics will determine if the proposed depthwise separable CNN architecture improves computational efficiency in the LithoBench framework while maintaining acceptable accuracy [1].

Detailed Project Execution Timeline

The project duration is 7 weeks, with development, experimentation, and report preparation occurring in parallel where appropriate.

Week 1-2 – Literature Review, Environment Setup & Baseline Model Implementation

- Review key research on NeuralILT, LithoBench, and efficient CNN architectures.
- Download and explore the LithoBench dataset and repository.
- Configure the development environment (Python, PyTorch, GPU on Google Colab).
- Implement dataset loading and verify layout–mask pairs.
- Begin drafting the Introduction and Related Work sections of the report.
- Implement the baseline U-Net encoder–decoder architecture used in NeuralILT experiments.
- Integrate the model with the MetalSet dataset loader.
- Implement the training pipeline (loss functions, optimizer, batching).
- Perform initial tests to verify correct training and inference behavior.

- Draft the Problem Statement and Proposed Solution sections.

Week 3 – Baseline Training and Initial Evaluation

- Train the baseline U-Net model on the MetalSet dataset.
- Monitor training loss and validation performance.
- Evaluate baseline metrics such as SSIM and Edge Placement Error (EPE).
- Record computational metrics including FLOPs, parameter count, and runtime.
- Document baseline model setup and training methodology.

Week 4 – DS-Conv Architecture Implementation

- Modify the baseline model by replacing standard convolution layers with depthwise separable convolution layers.
- Verify architectural correctness and parameter reduction.
- Integrate the modified architecture with the existing training pipeline.
- Begin drafting the Computational Cost and Architecture sections.

Week 5–6 – Parallel Experiments, Evaluation, and Documentation

Tasks during these weeks will run iteratively and partially in parallel:

- Train the DS-Conv NeuralILT model using the same training configuration as the baseline.
- Conduct controlled experiments comparing baseline and DS-Conv architectures.
- Evaluate performance using:
    - Edge Placement Error (EPE)
    - Structural Similarity Index (SSIM)
    - FLOPs
    - Parameter count
    - GPU inference runtime
    - Memory usage
- Evaluate generalization using the StdMetal dataset.
- Record intermediate results and update experiment documentation.

Week 7 – Analysis, Results, and Final Report

- Analyze experimental results and efficiency–accuracy trade-offs.
- Generate plots and tables comparing baseline and DS-Conv performance.
- Finalize the Results, Discussion, and Conclusion sections.
- Edit, review, and finalize the complete project report.

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Parking Space /Scratch Pad

2- Title Idea: Accelerating Inverse Lithography Technology (ILT) UsingConvolutional Neural Networks on the LithoBench Dataset

OR

Dataset: [GitHub - shelljane/lithobench: LithoBench: Benchmarking AI Computational Lithography for Semiconductor Manufacturing · GitHub]

Introduction

As the semiconductor industry continues to scale to advanced technology nodes, the complexity of lithographic patterning has increased significantly. With feature sizes approaching and even becoming smaller than the wavelength of the light used in photolithography, distortions during the printing process become unavoidable. These distortions introduce deviations between the intended design patterns and the patterns that are ultimately printed on silicon wafers. To mitigate these effects, semiconductor manufacturers rely on correction techniques that modify mask layouts before fabrication.

One of the earliest and widely used techniques is Optical Proximity Correction (OPC). OPC attempts to compensate for optical distortions by modifying mask features so that the printed patterns more closely match the intended layout. Traditional OPC approaches include rule-based OPC (RBOPC) and model-based OPC (MBOPC). While these approaches work well for regular Manhattan geometries, they struggle to accurately correct more complex or curvilinear shapes that appear in modern chip designs.

To address these limitations, Inverse Lithography Technology (ILT) was introduced as a more advanced mask synthesis technique. Unlike OPC, which adjusts mask edges using predefined rules, ILT treats the mask as a pixel-based optimization problem. By modeling the lithography process and iteratively refining the mask pattern, ILT can generate curvilinear mask shapes that significantly improve the fidelity of the printed wafer patterns. However, ILT is computationally expensive because it requires repeated physics-based simulations during the optimization process.

Recent advances in machine learning have introduced NeuralILT, which integrates neural networks into the ILT optimization pipeline to accelerate mask synthesis. NeuralILT models can learn mappings between target layout patterns and optimized mask patterns, reducing the need for repeated physics-based simulations. Despite these improvements, many existing NeuralILT implementations rely on deep convolutional neural networks such as U-Net architectures, which contain large numbers of parameters and require significant computational resources during

training and inference. When these models are scaled to full-chip optimization, the computational cost and memory requirements remain a major challenge.

This project explores a more computationally efficient approach to NeuralILT by leveraging depthwise separable convolutions within a convolutional neural network architecture. By decomposing standard convolution operations into depthwise and pointwise convolutions, the proposed model aims to reduce the number of parameters and floating-point operations while preserving the accuracy of mask predictions. Using the LithoBench MetalSet dataset, the project evaluates whether efficient convolutional architectures can accelerate NeuralILT while maintaining the fidelity required for advanced semiconductor manufacturing.

Short_Version_Project_Proposal_Accelerating_ILT

Accelerating Inverse Lithography Technology Using Depthwise Separable CNNs

Evaluation on the LithoBench MetalSet Dataset

Abstract

As Moore's Law pushes the physical limits of semiconductor manufacturing and the industry advances toward 3 nm process nodes, optical lithography faces increasing challenges due to diffraction and optical proximity effects [3]. Traditional correction techniques such as Optical Proximity Correction (OPC) and iterative Inverse Lithography Technology (ILT) are widely used to compensate for these distortions, but these approaches can become computationally expensive for complex layouts and advanced technology nodes [4][5].

Recent advances in machine learning have introduced neural network–based approaches that accelerate mask optimization workflows. Benchmark frameworks such as LithoBench demonstrate that deep learning models can learn the mapping between target layouts and optimized mask patterns, reducing the need for repeated physics-based simulations [1]. However, the computational overhead and high parameter counts of standard U-Net architectures commonly used with the LithoBench dataset remain a challenge for large-scale deployment.

This project proposes a more computationally efficient approach to NeuralILT by incorporating depthwise separable convolutions into the convolutional neural network architecture. Depthwise separable convolution, introduced in efficient CNN architectures such as MobileNet, decomposes standard convolution operations into depthwise and pointwise convolutions to significantly reduce the number of parameters and floating-point operations [6].

Using the open-source LithoBench MetalSet dataset, the proposed model aims to reduce computational cost during mask prediction while maintaining the accuracy required for advanced semiconductor nodes. The performance of the proposed architecture will be compared against the standard U-Net NeuralILT baseline using evaluation metrics such as mean squared error (MSE), structural similarity (SSIM), and edge placement error (EPE). The expected outcome is a lightweight NeuralILT model that improves computational efficiency while preserving the fidelity of generated mask patterns.

Introduction

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This project explores a more computationally efficient approach to NeuralILT by leveraging depthwise separable convolutions within a convolutional neural network architecture. By decomposing standard convolution operations into depthwise and pointwise convolutions, the proposed model aims to reduce the number of parameters and floating-point operations while preserving the accuracy of mask predictions. Using the LithoBench MetalSet dataset, the project evaluates whether efficient convolutional architectures can accelerate NeuralILT while maintaining the fidelity required for advanced semiconductor manufacturing.

Problem Statement

Although NeuralILT has significantly accelerated the traditional inverse lithography optimization process, existing implementations remain computationally expensive when applied to large-scale semiconductor layouts [1]. The standard U-Net architecture commonly used with the LithoBench MetalSet dataset can efficiently process individual layout tiles, but scaling these models to full-chip optimization requires substantial computational resources and memory [1]. This computational burden arises from the large number of convolutional kernels and parameters required to capture the complex curvilinear mask patterns present in advanced lithography datasets.

As semiconductor manufacturing continues to move toward smaller technology nodes, the computational cost associated with inverse lithography techniques becomes an increasingly important concern [3][5]. High parameter counts increase both memory consumption and inference time, making it difficult to deploy NeuralILT models efficiently in production-level mask synthesis workflows.

Therefore, there is a need to investigate more computationally efficient neural network architectures that can maintain mask prediction accuracy while reducing model complexity and computational overhead. This project aims to address this challenge by exploring the use of depthwise separable convolutions to redesign the NeuralILT architecture and evaluate whether such an approach can reduce computational cost while preserving mask fidelity on the LithoBench MetalSet dataset.

Related Work

Computational lithography research has explored various approaches to balance modeling accuracy and computational efficiency in mask optimization workflows [7]. Traditional lithography modeling techniques rely on detailed physical simulations of optical imaging, resist exposure, and etching processes. While these models can achieve high accuracy, they are computationally expensive and difficult to scale to full-chip optimization problems. As a result, modern research in computational lithography increasingly focuses on balancing physical modeling accuracy with computational efficiency [7].

Early machine learning methods, such as Support Vector Machines (SVMs) and decision tree models, were applied to tasks including lithography hotspot detection and EUV mask defect analysis [8,9]. These data-driven models could approximate complex lithographic behaviors without explicitly solving the underlying physical equations. However, their limited representational capacity restricted their effectiveness for modeling highly nonlinear optical proximity effects.

Deep learning has significantly expanded the capabilities of machine learning in lithography modeling. Convolutional neural networks (CNNs) and fully convolution networks (FCNs) can automatically extract spatial features from large layout datasets and approximate near-field diffraction patterns and aerial images with accuracy comparable to rigorous electromagnetic simulations while greatly reducing computational cost [10–12]. Generative deep learning methods such as LithoGAN and TEMPO accelerated mask optimization,achieving simulation speedups of several orders of magnitude compared with traditional rigorous solvers [13,14]. Similar approaches have also been applied to photoresist and etching process modeling, where machine learning models can approximate complex chemical and physical interactions during the fabrication process [15,16].

Deep learning has significantly accelerated Inverse Lithography Technology (ILT). Traditional pixel-based OPC methods require hundreds of iterative optimization steps to converge to an optimal mask solution. Deep learning models, including CNN-based architectures and generative models, have been proposed to directly learn the nonlinear mapping between target layouts and optimized mask patterns, thereby significantly reducing computational runtime [17,18].

More recently, researchers have explored **model-driven deep learning (MDL)** approaches that combine physical lithography models with neural network architectures. By incorporating prior knowledge of imaging physics into neural network design, these approaches improve both interpretability and reliability of learned models while maintaining high computational efficiency [19–21].

While these deep learning approaches significantly accelerate computational lithography workflows, many existing models rely on large CNN architectures that introduce substantial computational overhead. This creates a trade-off between model accuracy and computational efficiency. This work builds on these developments by investigating whether **depthwise separable convolutional architectures** can reduce computational complexity in NeuralILT models while preserving mask prediction fidelity when applied to the LithoBench MetalSet benchmark dataset.

Proposed Solution

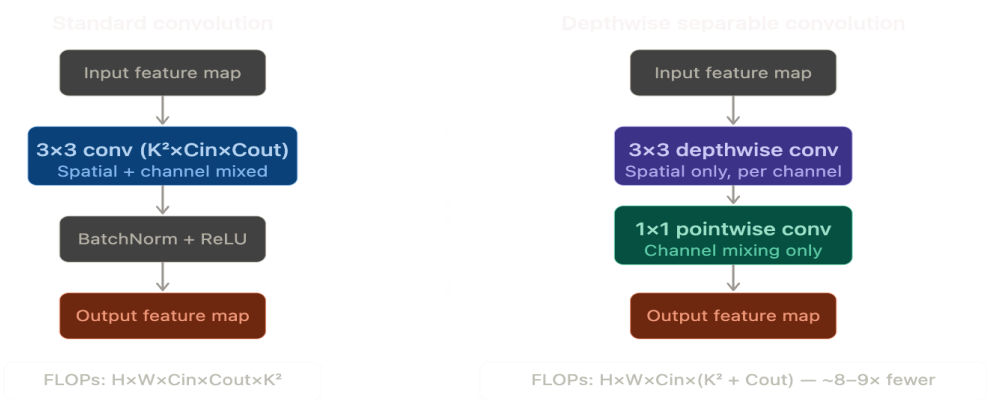
This project proposes implementing a fast and lightweight NeuralILT architecture using modified convolutional neural networks with depthwise separable convolutions on the LithoBench MetalSet dataset. The goal of this approach is to reduce computational cost and model complexity while maintaining acceptable mask prediction accuracy for advanced semiconductor manufacturing nodes.

In the LithoBench framework, neural networks are used to learn a mapping between the target layout pattern and the optimized mask pattern generated by inverse lithography technology (ILT) [1]. Current implementations commonly use encoder–decoder architectures similar to U-Net, which contain a large number of convolutional layers and parameters [1]. While these models can achieve good mask prediction accuracy, their computational requirements become significant when applied to large-scale datasets or full-chip optimization tasks.

This project proposes modifying the convolutional layers used in the baseline NeuralILT architecture by replacing standard convolution operations with depthwise separable convolutions. This modification decomposes a standard convolution into two separate operations: depthwise convolution and pointwise convolution.

In a standard convolutional neural network layer, spatial filtering and channel mixing are performed simultaneously in a single operation, which requires a large number of parameters and computational resources. Depthwise separable convolution divides this process into two stages. In the depthwise convolution, a small spatial filter is applied independently to each input channel. This operation captures local spatial information for each feature map. In the pointwise convolution, a (1×1) convolution is used to combine information across channels, allowing the model to learn interactions between different feature maps.

By separating these two operations, depthwise separable convolutions significantly reduce computational complexity by a factor of 8-9 compared to standard convolution layers while preserving the representational capacity of the network. Efficient convolutional architectures such as MobileNet demonstrate that depthwise separable convolution can greatly reduce the number of parameters and floating-point operations (FLOPs) required for deep learning models [6]. This architectural modification enables the development of a more scalable NeuralILT model that can be applied to complex lithography datasets such as LithoBench MetalSet.



Computational Cost of Convolution Operations

To motivate the proposed architectural modification, we compare the computational complexity of standard convolution and depthwise separable convolution, measured using floating-point operations (FLOPs).

Standard Convolution FLOPs

For a standard convolution layer, the number of floating-point operations is:

$$FLOPs_{std} = H \times W \times C_{in} \times C_{out} \times K^2$$

Where:

- (H, W) = height and width of the feature map
- (C_{in}) = number of input channels
- (C_{out}) = number of output channels
- (K) = kernel size (e.g., ($K = 3$) for a (3×3) convolution)

In this operation, spatial filtering and channel mixing occur simultaneously, which leads to a large number of parameters and high computational cost.

Depthwise Convolution FLOPs

Depthwise convolution applies a spatial filter independently to each input channel:

$$FLOPs_{dw} = H \times W \times C_{in} \times K^2$$

This operation captures local spatial patterns in each feature map while avoiding channel mixing.

Pointwise Convolution FLOPs

Pointwise convolution uses a (1×1) kernel to combine information across channels:

$$FLOPs_{pw} = H \times W \times C_{in} \times C_{out}$$

This stage performs channel mixing after the spatial filtering step.

Total FLOPs for Depthwise Separable Convolution

Combining both operations results in the total computational cost:

$$FLOPs_{sep} = H \times W \times (C_{in} \times K^2 + C_{in} \times C_{out})$$

Compared with standard convolution, depthwise separable convolution significantly reduces computational cost while preserving spatial feature extraction and channel interaction capabilities. This property makes it suitable for improving the efficiency of NeuralILT architectures applied to the LithoBench dataset.

In other words, for a 3x3 kernel with Cout = 64, the theoretical FLOPs reduction ratio is about 1 / (1/64 + 1/9) ≈ 8.1 x, which verifies that the proposed architecture is appropriate for large-scale mask synthesis tasks.

Expected Outcomes and Hypothesis

Based on the theoretical properties of depthwise separable convolution operation and the results of other lightweight CNN architectures such as MobileNet, the following outcomes are expected for the proposed NeuralILT architecture.

Computational efficiency: The DS-Conv architecture is anticipated to result in a considerable reduction in FLOPs and total parameters compared to the standard U-Net baseline and hence is more likely to be computationally efficient for deployment in large-scale mask synthesis workflows.

Inference efficiency: A faster inference time on GPU is anticipated due to the reduced FLOPs, particularly when performing a batch inference on a large number of layout tiles. A 2–4 times wall-clock time improvement is hypothesized, accounting for possible memory bandwidth effects that might offset the theoretical FLOPs benefits to some extent.

Mask prediction accuracy. Based on the performance of MobileNet and other similar architectures used in dense prediction tasks, it is expected that depthwise separable convolutions will retain a significant portion of the expressiveness of regular convolutions. It is hypothesized that the proposed model will be able to attain a similar or very close SSIM score to that of the baseline, i.e., within a range of 2-5%, and a similar or better EPE score within a certain tolerance range to that of the LithoBench benchmark.

Generalization. Due to the reduction in parameters of the DS-Conv model, which reduces the risk of overfitting to the MetalSet distribution, it is hypothesized that the proposed model will generalize similarly to or better than the baseline model over the StdMetal out-of-distribution evaluation set. These hypotheses will be verified through a series of experiments, which are presented in the following section. The evaluation will be performed using the evaluation metrics presented in the Proposed Evaluation Metrics section.

Training Strategy

The proposed model will be trained using a supervised learning approach with the ground truth mask patterns provided in the LithoBench dataset. In this framework, the neural network learns a mapping between the target layout pattern and the corresponding optimized mask pattern generated through inverse lithography technology (ILT) [1].

The training process will use the MetalSet subset of the LithoBench dataset, which contains complex metal routing patterns with strong optical proximity effects. Each input sample consists of a target layout tile, while the corresponding ground truth mask image serves as the training

label. The model will learn to predict optimized mask patterns that closely match the ground truth masks.

During training, the predicted mask output generated by the neural network will be compared to the ground truth mask using pixel-level loss functions. Mean Squared Error (MSE) will be used to measure the difference between predicted and ground truth mask intensities across all pixels.

$$L_{MSE} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2$$

Where:

- (y_i) represents the ground truth mask value
- ($\hat{y}_i$) represents the predicted mask value
- (N) represents the total number of pixels in the image

In addition to pixel-wise loss, the Structural Similarity Index (SSIM) will be used as a complementary evaluation metric to assess structural similarity between predicted and ground truth masks. SSIM measures similarities in local image structures such as edges, lines, and corners, which are critical for maintaining the fidelity of lithography mask patterns.

The training process will be implemented using the PyTorch deep learning framework and executed on a GPU-enabled environment. The Adam optimizer will be used for gradient-based optimization with mini-batch training.

To ensure a fair comparison, the proposed depthwise separable convolution architecture will be trained using the same dataset and training configuration as the baseline NeuralILT-style convolutional model.

After training on the MetalSet dataset, the model will be evaluated on both the MetalSet testing subset and the StdMetal dataset to measure generalization performance.

Proposed List of Experiments

The goal of the experiments is to evaluate whether replacing standard convolution layers with depthwise separable convolutions can improve the computational efficiency of NeuralILT models while maintaining mask prediction accuracy.

Experiment 1: Our Baseline Model- NeuralILT Model

The first experiment establishes a baseline using a standard convolutional neural network architecture similar to the NeuralILT U-Net model used in LithoBench benchmarks [1].

- Dataset: LithoBench MetalSet
- Input: Target layout tiles

- Output: Optimized mask tiles
- Training: Supervised learning using ground truth masks

This experiment provides baseline performance for mask prediction accuracy and computational cost.

Experiment 2: Our Solution Model- Depthwise Separable Convolution Model

In the second experiment, the standard convolution layers used in the baseline architecture will be replaced with depthwise separable convolution layers.

- Dataset: LithoBench MetalSet
- Training configuration identical to Experiment 1
- Same optimizer, dataset split, and training parameters

This experiment evaluates whether the proposed architecture can reduce computational complexity while maintaining similar mask prediction accuracy.

Experiment 3: Performance Comparison

The results from the baseline model and the proposed model will be compared to measure improvements in both prediction quality and computational efficiency.

Mask Fidelity Metrics

- Edge Placement Error (EPE)
  Measures the distance between predicted mask edges and the ground truth mask edges.
- Structural Similarity Index (SSIM)
  Measures similarity between predicted and ground truth mask images.

Computational Efficiency Metrics

- **FLOPs** – number of floating-point operations required during inference
- **Parameter Count** – model complexity and memory footprint
- **Inference Runtime** – time required to generate mask predictions
- **Memory Usage** – GPU memory consumption during inference

Experiment 4: Generalization Evaluation

To evaluate model robustness, both models will also be tested on the StdMetal dataset, which contains layouts that differ from the MetalSet training distribution [1].

This experiment evaluates how well the proposed architecture generalizes to new lithography patterns.

Experimental Objective

The experiments aim to determine whether the proposed depthwise separable convolution architecture can:

- Reduce computational cost
- Reduce model complexity
- Maintain acceptable mask prediction accuracy
- Improve scalability of NeuralILT models for practical lithography workflows

Experiment Setup and Dataset Details

LithoBench Dataset

The experiments in this project will be conducted using the LithoBench dataset, an open-source benchmark designed for evaluating machine learning approaches in computational lithography. The dataset provides layout–mask pairs used for training and evaluating neural networks for inverse lithography technology (ILT).

The LithoBench dataset is derived from the ICCAD-13 CAD contest benchmarks, which define design rules for a specific semiconductor technology node. These benchmarks simulate realistic optical proximity effects encountered in semiconductor manufacturing.

The dataset contains over 120,000 layout tiles and is divided into four subsets:

Dataset	Number of Tiles	Description
MetalSet	16,472	Complex metal routing patterns with strong optical proximity effects
ViaSet	116,415	Simpler via patterns with relatively isolated structures
StdMetal	271	Standard metal layouts used for evaluating model generalization
StdContact	328	Standard contact patterns used for cross-domain testing

In this project, MetalSet will be used as the primary dataset for training the NeuralILT models because it contains complex optical proximity effects that make mask prediction more challenging.

The StdMetal dataset will be used to evaluate model generalization to unseen layouts, while StdContact may be used for additional cross-domain testing. The ViaSet subset provides simpler lithography patterns that can help analyze how models behave under less complex optical conditions.

The dataset is distributed through the LithoBench GitHub repository and is available under the MIT License, allowing open-source research and experimentation.

Data Preprocessing Pipeline

Before training, all raw tiles from the LithoBench dataset undergo a consistent preprocessing pipeline to ensure data quality and a fair comparison between the baseline and proposed models.

Normalization. Pixel intensities for both layout input tiles and ground truth mask labels are normalized to the range [0, 1] by dividing by 255. This ensures stable gradient flow and consistent convergence behavior during training.

Dataset split. The 16,472 tiles of the MetalSet dataset are split into a training set, a validation set, and a test set, each taking 80%, 10%, and 10% of the data, which translates into 13,178, 1,647, and 1,647 tiles, respectively. This split is done with a fixed random seed to ensure reproducibility of the results in different experiments. The StdMetal set, which consists of 271 tiles, and the StdContact set, which consists of 328 tiles, are completely excluded from the dataset split and are only used during the evaluation phase.

Data augmentation. Horizontal flips, vertical flips, and 90-degree rotations are stochastically applied to the data during the training phase. These transformations are physically valid because mirror symmetry and rotational symmetry are geometrically meaningful in the context of lithography layouts. No photometric transformations such as brightness or contrast jittering are performed because the pixel values have a direct physical significance.

Tile handling and batching. The LithoBench dataset tiles are already pre-cropped to a size of 256 pixels. Therefore, no further processing is necessary. The tiles are loaded as single-channel grayscale images using PyTorch's Dataset and DataLoader utilities. The shuffle flag is set to True during training and False during validation/testing.

To make a fair comparison between the proposed DS-Conv model and the baseline U-Net model, both models are trained and tested using the same data processing pipeline.

Environment

The experiments will be conducted using a GPU-enabled environment to accelerate neural network training and inference.

Hardware

- Google Colab
- GPU: NVIDIA T4 or NVIDIA G4

These GPUs provide sufficient compute capability for training convolutional neural networks used in NeuralILT models.

Software Stack

The following software tools and libraries will be used:

- Python 3
- PyTorch – deep learning framework for model training and inference
- NumPy – numerical computation
- SciPy – scientific computing utilities
- scikit-image – used for computing image-based evaluation metrics such as Structural Similarity Index (SSIM)

This environment supports efficient training of both the baseline NeuralILT model and the proposed depthwise separable convolution model while allowing accurate evaluation of lithography mask prediction performance.

Proposed Evaluation Metrics

The performance of the proposed model will be evaluated using both **accuracy metrics** and **computational efficiency metrics**. These metrics allow comparison between the baseline NeuralILT model and the proposed depthwise separable convolution architecture.

Accuracy Metrics

Edge Placement Error (EPE)
Edge Placement Error measures the distance between predicted mask edges and the ground truth mask edges. This metric is widely used in computational lithography to evaluate how accurately a predicted mask reproduces the intended layout pattern [1].

Lower EPE values indicate that the predicted mask edges are closer to the desired lithographic pattern.

Structural Similarity Index (SSIM)
SSIM measures the similarity between predicted mask images and ground truth masks based on structural information such as edges, contrast, and spatial patterns. Unlike pixel-wise error metrics, SSIM evaluates perceptual similarity between images and is useful for assessing mask fidelity.

SSIM values range between:
- **1** → identical images
- **0** → no structural similarity

Higher SSIM values indicate better structural similarity between predicted and ground truth masks.

Computational Efficiency Metrics

Parameter Count
The total number of trainable parameters in the model will be measured to evaluate model complexity. Depthwise separable convolution is expected to significantly reduce the number of parameters compared to standard convolution layers.

Floating Point Operations (FLOPs)
FLOPs measure the total number of floating-point operations required during inference. This metric provides an estimate of the computational cost of the model. The proposed architecture aims to reduce FLOPs by replacing standard convolution layers with depthwise separable convolution layers.

Inference Runtime
Inference runtime measures the time required for the model to generate mask predictions for a given layout tile on the GPU.

Memory Usage
GPU memory usage during inference will also be measured to evaluate the memory efficiency of the proposed architecture.

These metrics will be used to evaluate whether the proposed depthwise separable convolution architecture improves computational efficiency while maintaining acceptable mask prediction accuracy within the LithoBench benchmarking framework [1].

Detailed Project Execution Timeline

The project duration is 7 weeks, the following timeline is proposed:
Week 1:Setup of the dataset, environment config, literature review
Week 2-3:Baseline implementation of the U-Net model & training
Week 4-5:Implementation & training of the DS-Conv model

Week 5-6:Experiments, metric collection
Week 7:Analysis,writing report

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